Keerthana Gandhirajan

Gen AI & Agentic AI Engineer | LLM & RAG | Python | Chennai, India
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Portfolio: keerthanagandhi11.github.io/keerthana-portfolio

Summary

Gen AI engineer building RAG systems, LLM agents and AI ops automation in Python. Shipped OpsPilot (agentic RAG with cited, self-checked answers) and OnCallPilot (AI on-call escalation). At Kyndryl: workload automation, mainframe-to-UAC migration.

Projects

OpsPilot: Agentic RAG for Incident PDFs | *Python, Groq, FastAPI, Gradio, Docker*
- Built a gpt-oss-120b agent that decides what to search, calls 4 tools over up to 3 rounds and cites file and page
- Combined BM25 keyword search with bge-small embeddings using Reciprocal Rank Fusion; the right page was in the top 3 results for 18/18 test questions (17/18 with keywords alone)
- Added hallucination guardrails: a second LLM call checks each claim against the cited pages, and the agent says "not found" instead of guessing (refused 5/5 unanswerable questions)
- Scored 18/18 correct answers with the right page cited in 18/18 on a 23-question eval set I wrote; median response time 2.2 s
- Deployed on Render with Docker and FastAPI, with rate limiting, GitHub Actions CI and 72 automated tests

OnCallPilot: AI On-Call Escalation for UAC Failures | *Python, FastAPI, SQLAlchemy, Groq, ServiceNow, Teams*
- Automated the manual UAC escalation: Teams alert, ServiceNow on-call lookup, call ladder, DL email, ticket transfer
- gpt-oss-120b triages secret-masked job logs into schema-validated JSON (rule fallback); a vision LLM reads screenshots
- A database-backed state machine, not the LLM, decides who to call and when; survives restarts and groups failure storms
- Deployed on Render with Docker and CI; 70 automated tests and an end-to-end run of 8 escalation scenarios (8/8 pass)

IT Operations AI Agent | *Python, LangGraph, Claude API, Model Context Protocol (MCP)*
- Built a LangGraph workflow on the Claude API that routes each ops request to the right tool via function calling
- Wrote 2 MCP tools (job-status checks, issue triage) and split the work across subagents, one per step

Additional Projects | *Python, LangChain, Vector Database, REST APIs, SQL, NLP*
- **RAG PDF Chatbot:** LangChain retrieval over embedded PDF chunks in a vector DB; Claude answers only from them
- **Job Notification API:** REST backend that sends real-time job alerts filtered by role, location and keywords
- **X Sentiment Analysis:** pulled posts from the X API, cleaned the text and labeled each one positive, negative or neutral

Experience

Kyndryl Jun 2025 – Present
Computer Operations Representative *Full-time, On-site*
- Helped migrate legacy mainframe batch jobs to Stonebranch UAC, checking job definitions, schedules and dependencies
- Worked on automated job workflows in UAC that cut manual steps out of recurring batch runs
- Monitored enterprise workloads, fixed failed jobs using runbooks and escalated incidents to the right support team

Technical Skills

Languages & CS: Python, Java, SQL, Object-Oriented Programming, Data Structures & Algorithms, Git/GitHub
Gen AI & LLMs: Generative AI, Multimodal AI, OpenAI API, Claude API, Hugging Face, Fine-Tuning, Prompt Engineering
Agentic AI: Tool Calling, Multi-Agent Systems, Subagents, LangGraph, MCP, Google ADK, CrewAI, AutoGen
RAG & Vector DBs: Agentic RAG, Hybrid Search, Embeddings, LangChain, LlamaIndex, ChromaDB, Pinecone, FAISS
Evaluation & Safety: RAGAS, Eval Sets, LLM-as-Judge, Guardrails, Hallucination Mitigation, LangSmith (Observability)
ML & Data: Machine Learning, NLP, PyTorch, TensorFlow, PostgreSQL, SQLAlchemy, MongoDB, Data Visualization
Backend & Cloud: FastAPI, Flask, Django, REST APIs, Gradio, Streamlit, Docker, AWS, GitHub Actions CI/CD, pytest
Automation & Ops: Stonebranch UAC, ServiceNow, Teams/Graph API, Workload Automation, Mainframe Migration, n8n, Incident Triage

Education

Bharathidasan University Tiruchirappalli, India
Master of Computer Applications (MCA) *Jul 2025 – Jul 2027 (expected)*

SASTRA Thanjavur, India
Bachelor of Computer Applications (BCA) *Sep 2022 – Jul 2025*

Certifications

Kyndryl AI Agent OPS Specialist Program (manager-nominated, in progress)
AI at Kyndryl: Mainframe Modernization
IBM RAG & Agentic AI Certification (in progress) | **Meta Back-End Developer Professional Certificate** (2024)